

Day 02: Types of Data

Understanding Data Before Analysis

100 Days of Data Science

Why Understanding Data Types is Important

Before applying any statistical method or machine learning algorithm, we must first understand the type of data we are working with.

Choosing the wrong method for a data type can lead to:

- Incorrect analysis
- Misleading visualizations
- Poor model performance

Understanding data types helps us decide:

- Which statistical measures to use
- Which graphs to draw
- Which machine learning models are suitable

Main Types of Data

Data can be broadly classified into two main categories:

- Categorical (Qualitative) Data
- Numerical (Quantitative) Data

Categorical (Qualitative) Data

Categorical data represents characteristics or labels rather than numerical values. It describes **what type** of data something is, not **how much**.

Key points:

- Values are categories or labels
- Arithmetic operations are not meaningful
- Often summarized using counts or percentages

Examples:

- Gender
- Blood group
- Country name

Nominal Data

Nominal data is categorical data with **no natural order**.

Characteristics:

- Categories exist
- No ranking or order

Examples:

- Gender (Male, Female)
- Blood Group (A, B, AB, O)
- Nationality

Important intuition: There is no concept of greater or smaller in nominal data.

Ordinal Data

Ordinal data is categorical data with a **meaningful order**, but the difference between categories cannot be measured.

Characteristics:

- Categories exist
- Order is present
- Exact difference is unknown

Examples:

- Ratings (Poor, Average, Good, Excellent)
- Education level (School, College, Postgraduate)
- Customer satisfaction levels

Key idea: We know the order, but not how much better one category is than another.

Numerical (Quantitative) Data

Numerical data represents quantities that can be measured or counted.

Key points:

- Values are numbers
- Arithmetic operations are meaningful
- Used in calculations and modeling

Examples:

- Age
- Height
- Salary

Discrete Data

Discrete data consists of **countable** numerical values.

Characteristics:

- Countable values
- Usually whole numbers
- Gaps exist between values

Examples:

- Number of students in a class
- Number of cars in a parking lot
- Number of defective items

Important intuition: Fractional values usually do not make sense for discrete data.

Continuous Data

Continuous data consists of **measurable** values that can take any value within a range.

Characteristics:

- Measured values
- Decimal values are possible
- Infinite possible values

Examples:

- Height
- Weight
- Temperature
- Time

Key idea: There are no gaps between possible values.

Why Data Type Matters in Data Science

Understanding data types helps in:

- Choosing correct visualizations
- Selecting appropriate statistical measures
- Deciding suitable machine learning algorithms

Examples:

- Categorical data is often used in classification
- Continuous data is commonly used in regression

Common Beginner Confusions

- Treating ordinal data as numerical
- Using mean for categorical data
- Confusing discrete and continuous data

Correct identification of data type avoids these mistakes.

Final Takeaway

Understanding data types is the first step before any analysis. If data type is misunderstood, even the best models can fail.

Good Data Science always starts with understanding the data.